

Curriculum Vitae

Personal Information

Name: George David Tsitlauri
Profile: AI & Systems Research Engineer
Date of Birth: 14 October 2003
email: gdtsitlauri@gmail.com

Education

2026-2027: Master in Artificial Intelligence & Applications, University of Thessaly, Greece.
2021-2026: Bachelor in Informatics & Telecommunications, University of Thessaly, Greece.
Specialization: Data Management and Artificial Intelligence
Research Laboratory: Computer Architecture, Compilers and System Security

Languages

Greek Native
Russian Native
English ECCE - University of Michigan

Research

Crypto3DStackCPU *Secure CPU prototype with a 4-layer 3D-stacked-memory abstraction.*
CUDA-zkML *GPU-accelerated zero-knowledge proofs for verifiable neural inference.*
Kronos *Causal-inspired generalization robust to unseen shifts and severe shortcut reversal.*
Nous *Self-improving LLM framework with reasoning loops for limited compute.*
PATV-X *Training-free, explainable deepfake-video detection via frequency analysis.*
Asclepius *Explainable multimodal biosignal AI for anomaly detection and federated health.*
Atlas *Simulated cloud autonomy with digital twins, orchestration and intelligent control.*
Olympus *AI security platform for threat simulation, reverse engineering, and malware analysis.*
Odyssey *Risk-aware intrusion-detection framework for post-quantum transition environments.*
Gate-6G *Adaptive 6G-style transmission via semantic communication and split inference.*

Academic Projects

CryptoCPU *(Thesis) Design and implementation of an encrypted instruction set processor on FPGA (Prof. G. Dimitriou).*
IR Optimizer *Intermediate-representation analysis and optimization for the CCC compiler.*
GPBFT *Group-based PBFT consensus evaluated in SymbChainSim.*
Autoscaler *Kubernetes weather app with autoscaling and load-aware deployment.*
Nexus *Compiler platform with MIPS generation, CPU simulation, parallelism and HDL tests.*

Websites

Skyrianos *Skyros island website with visitor information.*
OstrichZooPark *Ostrich Zoo Park website with visitor information.*
Pavlidis Home *E-commerce website for online product presentation and storefront workflows.*

Skills

Programming Python, C, C++, Java, SQL, JavaScript, TypeScript, Assembly, VHDL, Verilog, Go, C#, Scala, R, MATLAB
AI PyTorch, TensorFlow, XGBoost, Apache Spark, Apache Hadoop
Hardware CUDA, OpenMP, TensorRT, Xilinx Vivado
Web HTML5, CSS3, React, Next.js, Node.js, Tailwind CSS
Dev Git, CI/CD, Linux, Windows, Bash, CMake, Docker, Kubernetes, Wireshark, PostgreSQL, FastAPI, Jupyter, LaTeX